

→ Set is unordered collection of distinct elements.

→ It can be finite or infinite.

→ Null set: $\{\}$ or ϕ

→ For a set A , A and ϕ are trivial subsets.

→ If A is set (finite) then set of all subsets of A , called as power set $P(A)$.

$|A|$ = Cardinality of set.

$A = \{a, b\}$ $|A| = 2$

∴ power set include all subset $\{\phi, \{a\}, \{b\}, \{a, b\}\} \rightarrow P(A)$

$$|P(A)| = 2^{|A|}$$

$A - B$: elements belong to A only

$A \cap B = \phi$: Disjoint sets.

Symmetric difference / Boolean sum

$$A \Delta B / A \oplus B \checkmark \text{ not } \times$$
$$= (A - B) \cup (B - A)$$

→ sets are commutative and associative over $(\cap, \cup, \phi)$, and distributive over $(\cap, \cup)$.

$$A - (B \cap C) = (A - B) \cup (A - C)$$

Idempotent law

$$A \cup A = A \cap A = A \checkmark$$

Absorption law

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

$$\rightarrow A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Modular law

$$(1) (A \cup B) \cap C = A \cup (B \cap C) \text{ if } A \subseteq C$$

$$(2) (A \cap B) \cup C = A \cap (B \cup C) \text{ if } C \subseteq A$$

$$\rightarrow \checkmark A \cap P(A) = \phi$$

$$\rightarrow \checkmark P(A) \cap P(P(A)) = \{\phi\}$$

$$\rightarrow A \oplus (B \cup C) \neq (A \oplus B) \cup (B \oplus C)$$

$\oplus$ is not distributed over $(\cup \text{ or } \cap)$.

Relations

→ Also called as cross product.

Cartesian Product: Consider two sets.

$$A = \{1, 2, 3\} \quad B = \{a, b\}$$

$$A \times B = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$$

$$\text{If } |A| = n, |B| = m \text{ then } |A \times B| = m \times n$$

Relation is subset of Cartesian Product.

$$R = \{(1, a), (1, b)\} \text{ we can take any elements}$$

$\pm R$ can be said as $(1, a) \in R$

No. of finite relations over Cartesian product

equals to N ,

$$N = 2^{m \times n}$$

$$\text{Ex} \rightarrow R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x < y\}$$

$\mathbb{Z}^+ \rightarrow$ (ve) Integer $\mathbb{Z}^- \rightarrow$ negative integers

$$R = \{(1, 2), (2, 3), (3, 4), (1, 4), (1, 9), \dots\}$$

→ It is infinite relation.

→ Relation can be finite or infinite depends on nature of Cartesian product.

→ Relation can be defined on either two same sets ($A \times A$) or different sets ($A \times B$).

→ Reflexive Relation

A relation ' R ' on a set ' A ' is said to be reflexive if $(x, x) \in R \forall x \in A$.

$$\text{Ex } A = \{1, 2, 3\}$$

then R must contain $\{(1, 1), (2, 2), (3, 3)\}$

R can also take other these elements but these elements should be there.

→ If $|A| = n$ then smallest reflexive relation R_n contains ' n ' order pairs.

→ Largest reflexive relation contains all order pairs, $|R_n| = n^2$ (here).

→ If $A = \{1, 2, 3, \dots, n\}$ then no. of possible reflexive relation over A is equals to

$$|R_n| = 2^{(\text{No. of non-diagonal elements})}$$

$$= 2^{m \times n - n} = 2^{n(m-1)}$$

$$\text{So } A_n = 2^{n(n-1)} \quad [\because m=n \text{ here}]$$

No. of non-reflexive relation,

$$= 2^{n^2} - 2^{n^2 - n}$$

Note → ① The relation ' $\leq$ ' is reflexive on any set of real numbers. ($x \leq x \in \mathbb{R} \times \mathbb{R}$)

② The relation ' is a divisor of ' is reflexive on any set of non-zero real no. ($x/x \in \mathbb{R} \times \mathbb{R}$)

③ The relation ' is a subset of ' is reflexive on any collection of set. ($A \subseteq A \in A$)

④ The relation ' is parallel to ' is reflexive on a set of all lines.

⑤ $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x-y \text{ is even integer}\}$ $x-x=0$ (even).
→ ' $x \equiv x \pmod{5}$ ' on $\mathbb{Z}$ is reflexive

→ If R_1, R_2 are reflexive then,
 $R_1 \cap R_2 \rightarrow$ reflexive
 $R_1 \cup R_2 \rightarrow$ reflexive

Comparison on some set through of values

Ex = $A = \{1, 2, 3\}$

$R_1 = \{(1, 1), (2, 2), (3, 3), (3, 2), (1, 2)\}$

$R_2 = \{(1, 1), (2, 2), (3, 3), (2, 3), (2, 1)\}$

→ $R_1 \cap R_2$ and $R_1 \cup R_2$ both are reflexive

Irreflexive contains no diagonal elements. Even if a relation contain single diagonal element it will not be irreflexive.

Ex → $A = \{1, 2, 3\}$

$R_1 = \{(1, 1), (2, 2), (3, 3)\}$
↳ Reflexive

$R_2 = \emptyset \rightarrow$ Irreflexive

$R_3 = \{(2, 2)\} \rightarrow$ Neither reflexive nor irreflexive

$R_4 = \{(1, 2), (2, 1)\} \rightarrow$ Irreflexive but not reflexive

→ There is no relation which is both reflexive and irreflexive

→ If $|A| = n$ then,

→ Smallest irreflexive relation,
 $R_n = \emptyset, |R_n| = 0$

→ Largest irreflexive relation
 $R_n = A \times A - \text{d diagonal elements}$
 $|R_n| = n^2 - n$

→ Say $|A| = n$ and relation defined on $A \times A$ then no. of irreflexive relation -

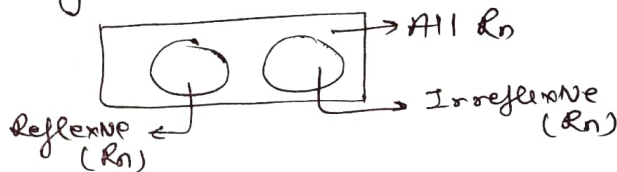
$A \times A = \{(1, 1), (2, 2), \dots, (n, n), (1, 2), (2, 1), \dots\}$

No. of diagonal element = n

Total element = n^2

No of irreflexive relation = $2^{n^2 - n}$

So, no. of reflexive relation equals to
no. of irreflexive relation = $2^{n^2 - n}$



$|R_n|$ which are either $R(R_n)$ or $I(R_n)$
 $= 2 \times 2^{n^2 - n} = 2^{n(n-1)+1}$

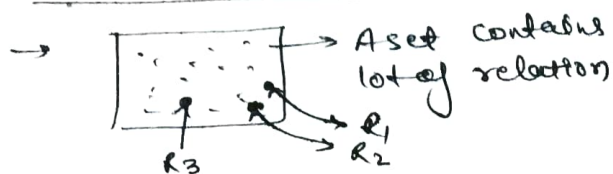
$|R_n|$ which are neither $R(R_n)$ nor $I(R_n)$
 $= 2^{n^2} - 2^{n(n-1)+1}$

Or $|R_n| = (2^n - 2) \times 2^{n^2 - n}$ for other elements
all possible relation with (x, x) because each set contains n (x, x) elements
subtract two cases that either includes all (x, x) elements or none of (x, x) elements

→ The relation ' $\leq$ ' on set of all real no., ' $\subseteq$ ' on set of all sets and ' $\perp$ ' on set of all lines are always irreflexive.

→ If R_1, R_2 are irreflexive then $R_1 \cup R_2$ and $R_1 \cap R_2$ are irreflexive

Closed Under operation



Say $R_1 \cup R_2 = R_3$

If R_3 belongs to same set then operation (op) can be said as closed. It means R_1 and R_2 are closed under operation (op).

Statement

→ The set of all reflexive relations are closed under the operation Set Union, Intersection only.

→ The set of all irreflexive relations are closed under the operation Set Union, Intersection and set difference.

→ If R_1, R_2 is reflexive then, $R_1 - R_2$ is irreflexive always.

Symmetric Relations

A relation 'R' on a set 'A' is said to be symmetric if $(xRy) \Rightarrow (yRx) \forall x, y \in A$.

$$A = \{1, 2, 3\}$$

$$R_1 = \{(1, 2), (2, 1)\} \rightarrow \text{Symmetric}$$

$$R_2 = \{(1, 1)\} \rightarrow \text{Symmetric}$$

$$R_3 = \{\} \rightarrow \text{Symmetric and Irreflexive}$$

$$R_4 = A \times A \rightarrow \text{Symmetric and Reflexive}$$

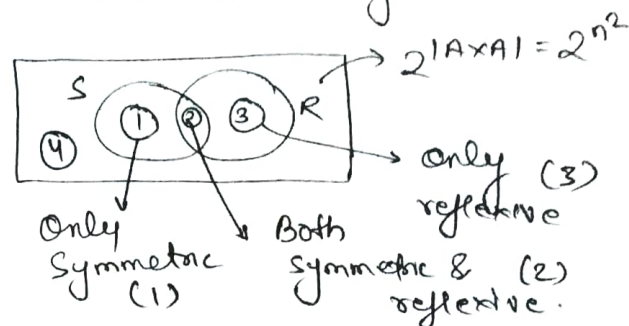
Smallest Symmetric relation R_3
 $|R_3| = 0$

Largest Symmetric relation R_4
 $|R_4| = n^2$

Relationship between symmetric and reflexive relations

$$|A| = n, \quad R \subseteq A \times A$$

Consider a Venn diagram as



→ Is it possible to form above diagram.

$$\text{Ex} \rightarrow A = \{1, 2, 3\}$$

$$R_1 = \{(1, 2), (2, 1)\} \rightarrow (1)$$

$$R_2 = \{(1, 1), (2, 2), (3, 3)\} \rightarrow (2)$$

$$R_3 = \{(1, 1), (2, 2), (3, 3), (1, 2)\} \rightarrow (3)$$

So, above diagram is valid.

$$|S| = 2^n \cdot 2^{\frac{n(n-1)}{2}} = 2^{\frac{n(n+1)}{2}}$$

$$|R| = 2^{n(n-1)}$$

Set of all relations which are both symm. and reflexive

$$A \times A = \{(1, 1), (2, 2), \dots, (n, n), (1, 2), (2, 1), \dots\}$$

$\underbrace{\hspace{10em}}_{n^2 - n}$

make pairs of $(n^2 - n)$ elements.

$$= 2^{\frac{n^2 - n}{2}} = 2^{\frac{n(n-1)}{2}}$$

$$\text{Region (3)} = 2^{\frac{n(n-1)}{2}} = |S \cap R|$$

$$|S - R| = n(S) - n(S \cap R) = 2^{\frac{n(n+1)}{2}} - 2^{\frac{n(n-1)}{2}}$$

$$|R - S| = n(R) - n(S \cap R) = 2^{n(n-1)} - 2^{\frac{n(n-1)}{2}} = 2^{\frac{n(n-1)}{2}} \left(1 - \frac{1}{\sqrt{2}}\right)$$

$$n(S \cup R) = 2^{n^2} - \left[n(S) + n(R) - n(S \cap R)\right]$$

If R_1 and R_2 are symmetric then $(R_1 \cap R_2)$, $(R_1 \cup R_2)$ and $(R_1 - R_2)$ are symmetric relations or closed relation.

No. of symmetric relation possible

$$|A| = n, (A \times A)$$

$$A = \{1, 2, 3\}$$

$$A \times A = \left\{ \begin{array}{ccc} (1,1) & (2,2) & (3,3) \\ \downarrow & \downarrow & \downarrow \\ (1,2) & (2,1) & \\ \downarrow & & \downarrow \\ (1,3) & (3,1) & (2,3) & (3,2) \end{array} \right\}$$

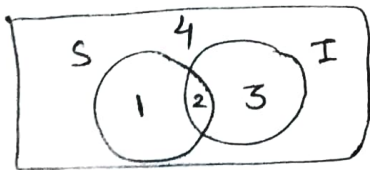
$$\text{No. of relation} = \boxed{2^n \times 2^{\frac{n^2-n}{2}}}$$

Symmetric

Here, $n^2 - n$ is an even number.

Relation b/w Symmetric and Irreflexive :

$$|A| = n \quad A \times A \rightarrow \text{set.}$$



We are trying to validate the above Venn diagram.

$$R_1 = \{(1,2), (2,1), (1,1)\} \rightarrow \textcircled{1}$$

$$R_2 = \{(1,2)\} \rightarrow \textcircled{3}$$

$$R_3 = \{(1,2), (2,1)\} \rightarrow \textcircled{2}$$

So, above diagram is possible

$$R_4 = \{(1,2), (1,1)\} \rightarrow \textcircled{4}$$

→ If gets failed then go for (Disjoint sets).

$$n(S) = 2^{\frac{n(n-1)}{2}} \cdot 2^n = \boxed{2^{\frac{n(n+1)}{2}}}$$

$$n(I) = \boxed{2^{\frac{n^2-n}{2}}}$$

$$n(S \cap I) = \boxed{2^{\frac{n^2-n}{2}}}$$

$$n(S \cup I) = n(S) + n(I) - n(S \cap I)$$

$$n(S - I) = n(S) - n(S \cap I)$$

$$n(I - S) = n(I) - n(S \cap I)$$

$$\overline{n(S \cup I)} = 2^{n^2} - n(S \cup I)$$

Antisymmetric Relations

A relⁿ is antisymmetric if (xRy) and (yRx) then $x=y$ $\forall$ $x, y \in R$

$$A = \{1, 2, 3\}$$

$$R_1 = \{(1,2), (2,1)\} \text{ Not Antisymmetric}$$

$$R_2 = \{(1,1)\} \text{ Symmetric, Antisymmetric}$$

$$R_3 = \{(1,2), (1,1)\} \text{ Antisymmetric}$$

$$R_4 = \{(2,3)\} \text{ Not Antisymmetric and not symmetric.}$$

$$R_5 = \{\} \rightarrow \text{Smallest antisymmetric Relation}$$

$$A \times A = \{(1,1), (2,2), (3,3),$$

$$\begin{array}{l} \boxed{(1,2), (2,1)} \quad \boxed{(3,1), (1,3)} \\ \boxed{(2,3), (3,2)} \end{array} \} \rightarrow \text{Pairs}$$

$$R_7 = \{(1,1), (2,2), (3,3), (1,2), (3,1), (3,2)\} \rightarrow \text{Largest Antisymmetric}$$

We can choose only one order pairs from pairs

If $|A| = n$ for $A \times A$

then cardinality for largest antisymmetric relation is -

$$|R_n| = n + \frac{n^2-n}{2} = \boxed{\frac{n(n+1)}{2}}$$

No. of antisymmetric relⁿ possible

$$|A| = n \quad |A \times A| = n^2$$

Ex $\rightarrow A = \{1, 2, 3\}$

AxA

$$= \frac{(1,1)}{2} \frac{(2,2)}{2} \frac{(3,3)}{2} \frac{(1,2)(2,1)}{3}$$

$$\frac{(1,3)(3,1)}{3} \frac{(2,3)(3,2)}{3}$$

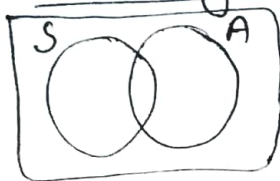
3 cases bcoz either include (1,3),
or (3,1) or non from them.

$$= 2^3 \times 3^3$$

for $|A| = n$

No. of antisymmetric relations
are = $2^n \times 3^{\frac{n^2-n}{2}}$

Relation between no. of symmetric
and antisymmetric relation.



$$|U| = 2^{n^2}$$

$$n(S) = 2^{\frac{n(n+1)}{2}}$$

$$n(A) = 2^n 3^{\frac{n(n-1)}{2}}$$

$$A = \{1, 2\}$$

$R = \{(1,1), (2,2)\}$ is a symmetric
as well as antisymmetric

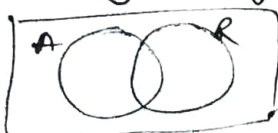
$$|R| = 2^n = n(S \cap A)$$

$$n(S \cup A) = n(S) + n(A) - n(S \cap A)$$

$$n(\overline{S \cap A}) = n(U) - n(S \cap A)$$

$$n(S - A) = n(S) - n(S \cap A)$$

Relation between reflexive and
antisymmetric



$$n(A) = 2^n 3^{\frac{n(n-1)}{2}}$$

$$n(R) = 2^{n(n-1)}$$

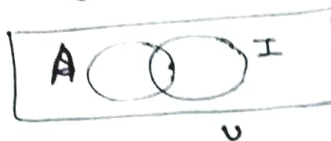
$$n(U) = 2^{n^2}$$

for make reflexive we need to
include all (x,x) element.
So they have no choice to exclude

$$n(A \cap R) = 3^{\frac{n(n-1)}{2}}$$

$$n(A \cup R) = n(A) + n(R) - n(A \cap R)$$

Relation b/w irreflexive and anti-
symmetric



$$|A| = n, A \times A$$

$$n(U) = 2^{n^2}$$

$$n(A) = 2^n 3^{\frac{n(n-1)}{2}}$$

$$n(I) = 2^{n(n-1)}$$

R is a relⁿ which is asymmetric
as well as irreflexive

$$R = \{(1,2)\}$$

$$n(A \cap I) = 3^{\frac{n(n-1)}{2}}$$

Now you can solve any query over it

The relation ' $\leq$ ' or ' $<$ ' is anti-
symmetric on any set of real
numbers.

The relation ' $\subseteq$ ' or ' $\subset$ ' (proper subset)
is also anti-symmetric
over any collection.

Asymmetric relation

$\rightarrow$ Stricter version of anti-symmetric.

$\rightarrow$ we are not allowing even (x,x) .

A relⁿ is asymmetric if (xRy) then
 $(yRx) \nexists x, y \in A$,

$$A = \{1, 2, 3\}$$

$R_1 = \{(1,2)\} \rightarrow$ Asymmetric and
Anti-symmetric.

$R_2 = \{(1,2), (2,2)\} \rightarrow$ Not asymmetric
but Asymmetric.

$\rightarrow$ Diagonal element can be present
in anti-symmetric but not in
asymmetric.

$R_3 = \{\} \rightarrow$ Asymmetric and
antisymmetric both.
 $\hookrightarrow$ Also symmetric too.

$R_4 = \{(1,2), (2,1)\} \rightarrow$ Not asymmetric
and antisymmetric.

Smallest = $0 : \{ \}$

Largest Asymmetric Relation

Largest = $\boxed{\frac{n^2 - n}{2}}$, $|A| = n$

No. of asymmetric relations

$|A| = n$, $A \times A \subset R$

$A = \{1, 2, \dots, n\}$ $|A \times A| = n^2$

$A \times A = \left\{ \begin{array}{l} \text{diagonal} + \text{non diagonal} \\ n \qquad \qquad \qquad (n^2 - n)/2 \\ \text{only 1} \qquad \qquad \text{include only} \\ \text{choice} \qquad \qquad \text{one from any} \\ \qquad \qquad \qquad \text{pair} \end{array} \right\}$

→ we can't include diagonal elements
→ we have 3 choices for non-diagonal

$|Asymmetr| = 3^{\frac{n(n-1)}{2}}$

→ if a relation is reflexive then it can't be asymmetric.

So,



$n(A) = 3^{n(n-1)/2}$

$n(R) = 2^{n(n-1)}$

$n(A \cup R) = n(A) + n(R)$

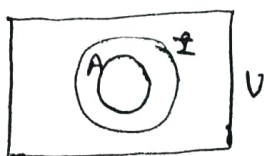
$n(A \cap R) = 0$

$n(A - R) = n(A)$, $n(R - A) = n(R)$

$n(\overline{R \cup A}) = n(U) - n(R \cup A)$

Relation between asymmetric and Irreflexive

→ Every asymmetric relation is Irreflexive, but converse is not true.



$n(I) = 2^{n(n-1)}$, $n(A) = 3^{n(n-1)/2}$

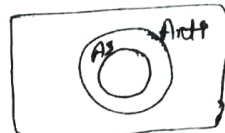
$n(I \cup A) = n(I)$ | $n(I - A) = n(I) - n(A)$

$n(I \cap A) = n(A)$ | $n(A - I) = 0$

$n(\overline{I \cup A}) = n(U) - n(I)$

Relation between Asymmetric and Antisymmetric

→ Every Asymmetric relation is antisymmetric



$n(A) = 3^{n(n-1)/2}$

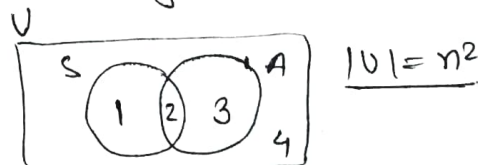
$n(Anti) = 2^n 3^{n(n-1)/2}$

$n(A \cup Anti) = n(Anti)$

Similarly we can find more values.

Relation between symmetric and Asymmetric

$R = \{ \}$ is symmetric and asymmetric both.



$R_1 = \{(1,1)\}$ | $R_2 = \{(1,2)\}$

$R_3 = \{ \}$ | $R_4 = \{(1,1), (1,2)\}$

↑
 $n(S \cap A) = 1$

$n(S) = 2^{n(n-1)/2}$

$n(A) = 3^{n(n-1)/2}$

$n(S \cup A) = n(S) + n(A) - 1$

$n(\overline{S \cup A}) = n(U) - n(S \cup A)$

Asymmetric relations are closed under subset, intersection and set difference operation.

Transitive Relations

If (xRy) and (yRz) , then
 $(xRz) \forall x, y, z \in A$.

If $a < b$ and $b < c$ then
 $a < c$. So it is transitive.

Similarly, $>$, $\leq$, $\geq$, $\neq$, $/$,
are transitive.

→ Closed under $\cap$ not $\cup$.

Equivalence relation

→ If the relation is reflexive,
symmetric and transitive then
it is called as equivalence
relation.

$R = \{ \text{diagonal elements only} \}$
is the smallest equivalence
relation.

$|R| = n$ (for smallest relⁿ)

$|R| = n^2$ (for largest relⁿ)

→ Closed under $\cap$ not $\cup$

Partial Order Relation (POR)

→ A relation on set 'A' is
POR if it follows all three
properties: reflexive, transitive
and antisymmetric together.

POSet → Partial Order Set

Ex of POR

$R = \{ (1,1), (2,2), (3,3) \}$

↳ Above relation is both POR
and equivalence relation.

✓ The set that holds a
partial order relation 'R' on
it is called as Poset.

→ It is denoted as $[A; R]$

Set $\swarrow$
POR

Ex: $[R; \leq]$, $[R; \geq]$, $[S; \subseteq]$,
 $[S; \supseteq]$, $[S; /]$

$R \rightarrow$ Set of all real no.

$S \rightarrow$ set of all sets.

$\supseteq \rightarrow$ Superset $\supset \rightarrow$ Proper superset

TOS (Totally Ordered Set)

Linearly ordered set or chain.

A Poset $[A; R]$ is called as 'TOS' if
every pair of elements in A are
comparable, i.e.; aRb or $bRa \forall$
 $a, b \in A$.

Ex → If 'A' is any set of real nos
then poset $[A; \leq]$ is a TOS.

Explanation: If you choose any two
nos from the set of real nos
you can apply ' $\leq$ ' symbol to
compare. Either both will be equal
or any one of them should be
smaller one.

② If $A = \{1, 2, 6, 30, 60, 300\}$ then
 $[A; /]$ is TOS.

→ In finding TOS order of
selection doesn't matter as
you can think 1 can divide
every no's similarly 2 can divide
all no's on RHS but how 2 can
divide 1. $\therefore$ we divide 1
by 2. It is not case we are just
choosing $(1,2)$ are applying division
as $(2/1)$ in both cases either
 $(1,2)$ or $(2,1)$.

Partial Orders and Lattice

→ Hasse diagram (Poset diagram)

Let $[A; R]$ be a poset. The poset diagram is as follows.

(1) There is a vertex corresponding to each element of A .

(2) An edge between the elements a and b is not present in the diagram, if there exists an element $x \in A$ such that (a, x) and (x, b) .

→ we don't represent any transitive relation.

(3) An edge b/w the elements a and b is present iff (a, b) and there no element $x \in A$ such that (a, x) and (x, b) .

Ex $A = \{1, 2, 3\}$ Relation: ' $\leq$ '

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$$

3. we can do this for

2. (1, 1) (2, 2) (3, 3) but

1. we know that it is poset and all posets

are reflexive. So, we have no need to make self loop in Hasse diagrams

So,



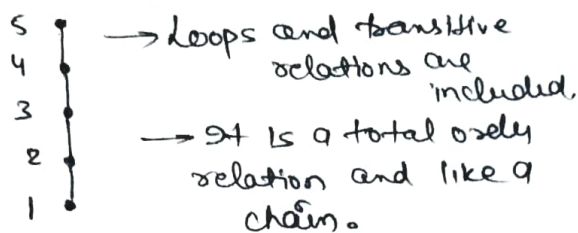
→ No need bcoz we can go from 1 to 3 using 2 already.

Also we have no need to mark the arrow bcoz it is already

understood from levels in diagram

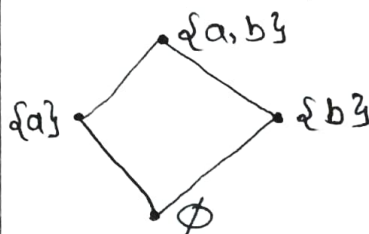
$\begin{matrix} 2 \\ 1 \end{matrix} \rightarrow$ 2 refers (1, 2) bcoz 2 at upper level in diag.

Ex → ① $A = \{1, 2, 3, 4, 5\}$ $[A; \leq]$

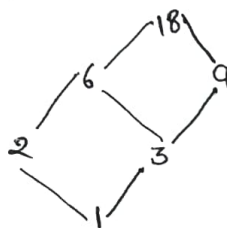


② $S = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

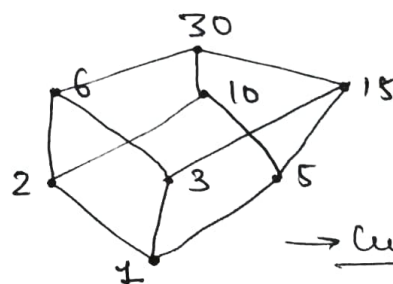
$S = P(A) : A = \{a, b\}$
Relation $[S, \subseteq]$



③ $A = \{1, 2, 3, 6, 9, 18\}$
 $[A, |]$



$A = \{1, 2, 3, 5, 6, 10, 15, 30\}$ $[A, |]$



→ Cube.

→ It is not necessary to always get a closed figure

LUB → Least Upper Bound
(LUB or join or supremum)

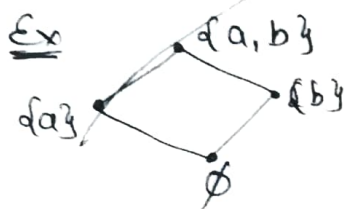
Let $[A; R]$ be a poset for $a, b \in A$, if there exist an

element $c \in A$ such that

(1) aRc and bRc

(2) if there exist any other element 'd' such that (aRd) and (bRd) then (cRd) , then c is called LUB of 'a' & 'b'.

Ex $\rightarrow (a, c)$ and $(b, c) : c$ is upper bound.



$(\{a\}, \{a, b\})$ $(\{b\}, \{a, b\})$
So, here, LUB = $\{a, b\}$

Similarly $(\emptyset, \{a\})$ $(\{a\}, \{a\})$
and $(\emptyset, \{b\})$ $(\{b\}, \{b\})$
So, $\{a\}$ and $\{b\}$ both are LUB. Now apply Rule 2 bcoz $(\{a\}, \{a, b\})$ $(\emptyset, \{a, b\})$ are also in the bucket.
So, $c = \{a\}$, $d = \{a, b\}$. $\exists n$ (cRd) LUB = $c = \{a\}$.

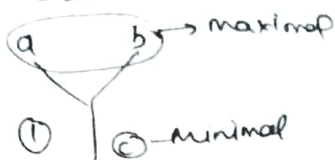
$\rightarrow$ So, $\exists n$ $(\{a\}, \{a, b\})$ and $(\{b\}, \{a, b\})$ and $(\emptyset, \{a, b\})$
LUB = $\{a, b\}$

$\rightarrow \exists n$ $(\emptyset, \{a\})$, $(\{a\}, \{a, b\})$
LUB = $\{a\}$

$\rightarrow \exists n$ $(\emptyset, \{b\})$, $(\{b\}, \{a, b\})$
LUB = $\{b\}$

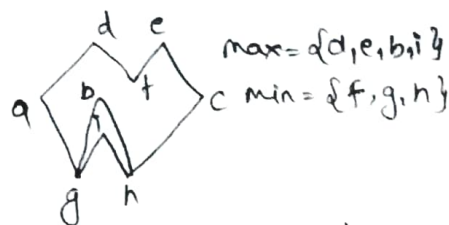
$\rightarrow$ Maximal, Minimal element

if in a poset, an element is not related to any other element, called maximal and if in a poset, no related element for an element called minimal



② c
 a b
maximal = $\{a, b, c\}$
minimal = $\{a, b, c\}$

$\rightarrow$ 2nd less diagram have reflexive property and also showing a poseto



$\rightarrow$ Each deep mesh have min. & maximal or minimal elements

$\rightarrow$ Maximum and minimum element

for maximum, it should be maximal and every elements related to it.

for minimum, it should be minimal and it is related to every element in poseto

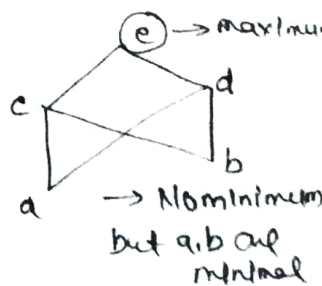
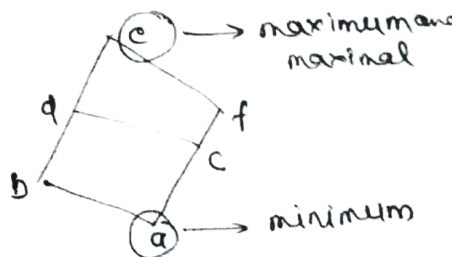


No maximum
d, e not related to each other

a, b are minimal but not minimum bcoz not related.

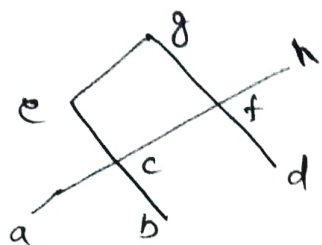
$\rightarrow$ Both maximum and minimum

a b c $\rightarrow$ No maximum and minimum



we need single bottom or single top

Upper Bound and Lower Bound



$$A = \{a, b, c, d, e, f, g\}$$

$$B = \{e, c\}$$

$$L.B(B) = \{a, b, c\}$$

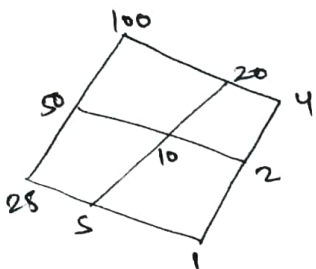
$$U.B(B) = \{g, e\}$$

We consider e, c in L.B and U.B because of self loops

$$B = \{c, f, d\}$$

$$L.B(B) = \phi$$

$$U.B(B) = \{f, g, h\}$$



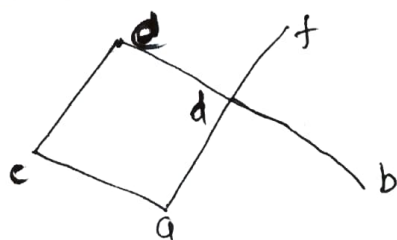
$$L.B(B) = \{1, 2, 5, 10\}$$

$$U.B(B) = \{100, 50\}$$

$$B = \{5, 10\}$$

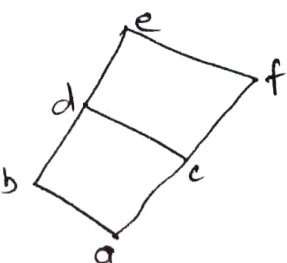
Least Upper Bound (Join, Supremum, $\vee$)

Greatest Lower Bound (Intimum, meet, $\wedge$)



$B = \{c, d\}$	$B = \{a, b\}$	$B = \{e, f\}$
$U.B(B) = \{e\}$	$U.B(B) = \{e, d, f\}$	$U.B(B) = \{f\}$
$L.B(B) = \{a\}$	$L.B(B) = \phi$	$L.B(B) = \{a, d, b\}$
$LUB(B) = \{e\}$	$LUB(B) = \{d\}$	$LUB(B) = \phi$
$GLB(B) = \{a\}$	$GLB(B) = \phi$	$GLB(B) = \{d\}$

$B = \{a, c, f\}$	$B = \{d, c\}$
$U.B(B) = ef$	$U.B(B) = e, d$
$LUB(B) = f$	$LUB(B) = d$
$L.B(B) = a$	$L.B(B) = ac$
$GLB(B) = a$	$GLB(B) = c$

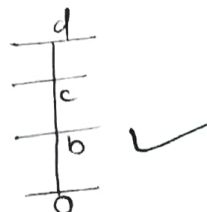
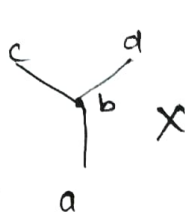


Join Semi-Lattice

→ We fix the number of element in the set as 2 for which we are finding UB and LB.

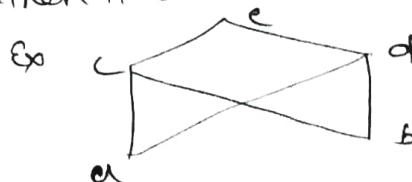
→ In a poset if LUB exist for every pair of elements, then poset is called as Join semi-lattice.

→ If you are able to find any single pair for which join doesn't exist then poset is not Join semi-lattice.

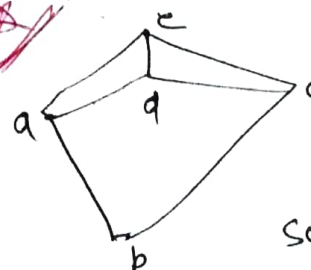


→ To find join for any pair find out the point where they meet firstly.

→ In case if there are more than 2 candidates for LUB then it will not be Join semi-lattice.



Ex c, d both are the candidates.



→ There are two candidates d, e but both are join.

So, we can take decision.

→ No single top → No Join semi-lattice

Meet semi-Lattice

In a poset if GLB/MEET/ $\wedge$ exists for every pair of elements then poset is called meet-semi lattice.

Lattice $\rightarrow$ If both LUB and GLB are exist for every pair of elements.

Ex $\rightarrow A = \{1, 2, 3, \dots, 10\}$ $[A, |]$ is a meet semi lattice.

$\rightarrow$ To verify take any pair and find its GCD, it will belong to A.

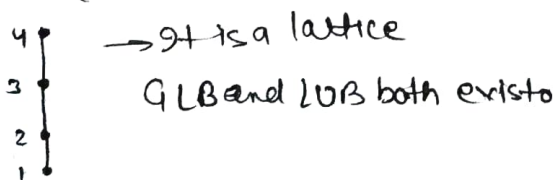
Here, GCD and GLB are same.

① $S = \{ \{a\}, \{b\}, \{a, b\} \}$ $[S, \subseteq]$ is a join semi-lattice.

$\rightarrow$ To verify you can find union of any two element means in case of subset Union acts as LUB.

$\rightarrow$ for GLB use Intersection.

③ $A = \{1, 2, 3, 4\}$ $[A, \leq]$



Properties of Lattice

for 3 elements in a lattice all these are true

① Commutative over $\wedge$ or $\vee$
 $a \vee b = b \vee a \rightarrow$ LUB
 $b \wedge a = a \wedge b \rightarrow$ GLB

② Associative

$$(a \vee b) \vee c = (b \vee c) \vee a$$
$$(a \wedge b) \wedge c = a \wedge (b \wedge c)$$

③ Idempotent law

$$a \vee a = a$$
$$a \wedge a = a$$

④ Absorption law

$$a \vee (a \wedge b) = a = a \wedge (a \vee b)$$

⑤ Distribution law doesn't hold for all lattice, only hold for distributive lattice.

$\rightarrow$ In a lattice $(a \vee b) \wedge c = b$ iff $(a \wedge b) = a \quad \forall a, b \in \text{Lattice}$

Distributive Lattice and Sub-Lattice

$$(1) a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$(2) a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

If a lattice satisfies above two properties called as distributive lattice.

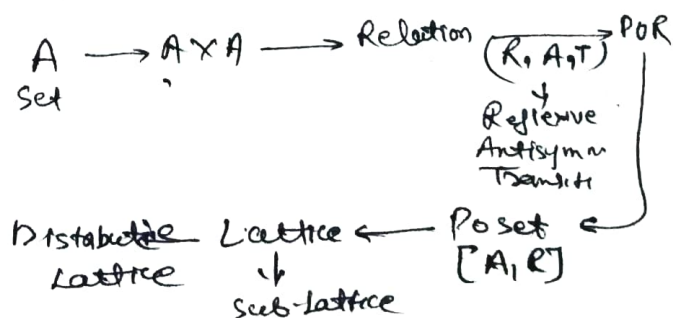
Sub-lattice:

Finding distributive lattice is a tedious task, so we use sub-lattice concept to reduce complexity.

Let 'L' be a lattice $[L, \vee, \wedge]$. A subset 'M' of 'L' is called a sublattice if

1) M is a lattice

2) for any pair of elements $a, b \in M$ the LUB and GLB are same in 'M' and 'L'.



On a bounded lattice, the following properties holds good.

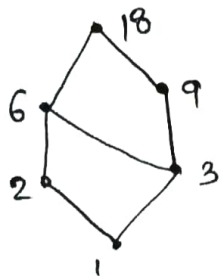
Say $I \rightarrow$ Upper bound
 $O \rightarrow$ lower bound.

- 1) LUB of a and I i.e. $a \vee I = I$
 - 2) GLB of a and I i.e. $a \wedge I = a$
 - 3) LUB of a and O i.e. $a \vee O = a$
 - 4) GLB of a and O i.e. $a \wedge O = O$
- (a, I)
 $\downarrow \quad \downarrow$
 GLB LUB

(O, a)
 $\downarrow \quad \downarrow$
 GLB LUB

$\rightarrow D_n =$ Set of all positive divisors of n .

Ex, $[D_{18}, /]$



Bounded Lattice

Lattice that have both lower and upper bound.

$\rightarrow$ On the given poset check for lattice, if and only if it has single top or single bottom

Upper bound of Lattice ' L ', if there exist an element I , such that $\forall a \in L, (a \leq I)$ then, I is called upper bound,

Lower bound of lattice ' O ', if all the elements in L have $(O \leq a) \forall a \in L$.

$\rightarrow$ for infinite lattice no upper and lower bound exist. Partially it exist but we can't find it

$\rightarrow$ for any set upper bound is Universal set and lower bound is ϕ .

$A \cup U = U$	$A \cup I = I$
$A \cap U = A$	$a \cap I = a$
$A \cup \phi = A$	$a \cup O = a$
$A \cap \phi = \phi$	$a \cap O = O$

$\rightarrow$ Bounded lattice must have lower or upper bound means finite lattice, not infinite

We know,

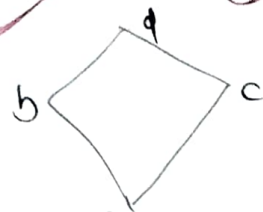
$A \cup A^c = U$	$a \cup a^c = I$
$A \cap A^c = \phi$	$a \cap a^c = O$

So, we can find complement of element using bounded lattices

Complement of an element in a lattice

In a bounded lattice ' L ' for any element $a \in L$, if there exist an element $b \in L$ such that $a \cup b = I, a \cap b = O$, that ' b ' is called complement of ' a ', we can say ' a ', ' b ' are complement of each others

$$I^c = 0, 0^c = I$$



$$a \vee d = d$$

$$a \wedge d = a$$

So, a, d are

$$I = d, 0 = a$$

complement

→ for 'v' go for first meet of both elements on the upper sides

→ for '∧' go for first meet of both elements on lower sides

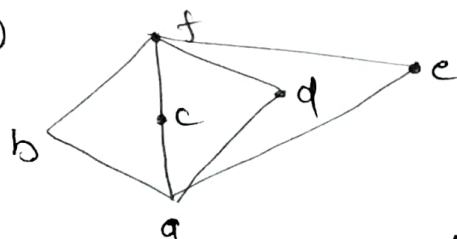
$$b \vee c = d$$

So, b, c are

$$b \wedge c = a$$

complement of each other

(2)

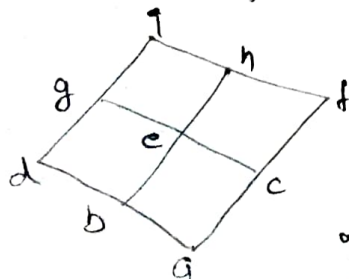


(a, f) are complement pair.

$$\begin{array}{l|l} b^c = c, d, e & c^c = b, c, d \\ c^c = b, d, e & d^c = b, c, e \end{array}$$

In set theory there should exist atleast and only one complement.

→ In lattice it is not necessary that there should exist a complement and there is need not be only one complements



(d, e) are not complement

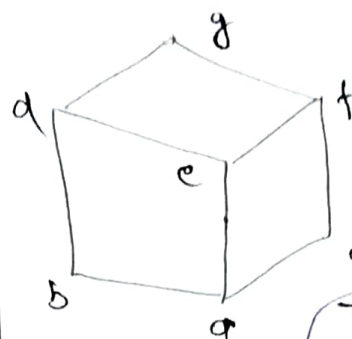
$$d \vee e = g \text{ not } f$$

$$d \wedge e = b \text{ not } a$$

$$i \rightarrow \text{GLB} \rightarrow I$$

$$a \rightarrow \text{LOB} \rightarrow 0$$

$a^c = i, d^c = f$, there is no Complement for b, c, e, g, h.



$$b^c = c, f$$

$$a^c = g, d^c = c$$

$$c^c = b, d$$

$$f^c = b$$

There is no complement for e.

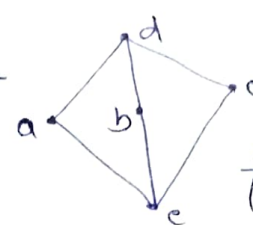
Distributive Lattice

A lattice is said to be distributive if $\forall a, b, c \in L$.

$$a) a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$b) a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

Ex



LHS

$$a \vee (b \wedge c)$$

$$a \vee c = a$$

RHS

$$(a \vee b) = d \quad | \quad d \wedge d = d$$

$$(a \vee c) = d \quad | \quad d \wedge d = d$$

$a \neq d$, so distributive law doesn't follow.

But checking property for each order pair of 3 elements is a tedious task. So we go for 9 tricks

→ In a distributive lattice, every element have atmost 1 Complement

In above diagram,

$$a^c = b, c \quad | \quad c^c = a, b$$

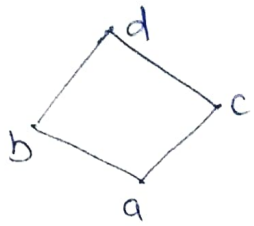
$$b^c = a, c$$

So, not a distributive lattice.

→ It is the easiest way to find distributive lattice.

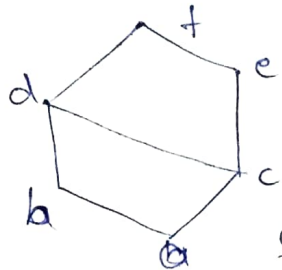
Complemented Lattice

✓ A lattice is said to be complemented if each element has atleast 1 complement



$$\begin{aligned} a^c &= d \\ b^c &= c \end{aligned}$$

Lattice is both distributive and complemented if each element must have only one complement

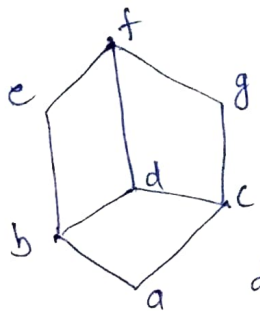


$$\begin{aligned} a^c &= f \\ b^c &= e \\ c^c &= d \end{aligned}$$

$a^c = \text{no complement}$

So it is not a complemented lattice.

→ If you have direct path then no need to go for other paths



$$\begin{aligned} a^c &= f, f^c = a \\ b^c &= g, c^c = e \\ c^c &= g, e^c = b \end{aligned}$$

$d \rightarrow$ no complement

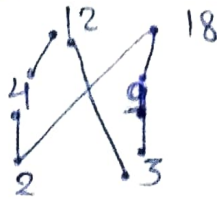
→ Not a complemented lattice

Boolean Algebra: A lattice 'L'

✓ is said to be boolean algebra, if it is complemented & distributive

→ Each element have exactly one complement

Ex $[2, 3, 4, 9, 12, 18, 1]$



Hasse diagram

→ It is not a lattice, just a Poset.

② $[R, \leq]$ $R \in (-\infty, \infty)$



→ Unbounded lattice

→ It is distributive but we will get no complement due to absence of Upper and lower bound.

③ $[D81, 1]$

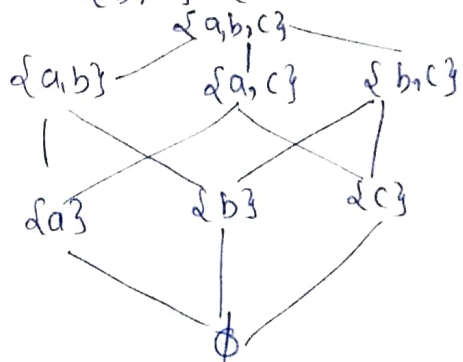
$$D = \{1, 3, 9, 27, 81\}$$



→ Bounded lattice but distributive and not complemented Lattice

④ $(P(A), \subseteq)$ $A = \{a, b, c\}$

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$$



Every $P(A)$ is boolean algebra w.r. to its sets

Groups

Algebraic Structure

A non-empty set 'S' is called an algebraic structure w.r.to binary operation 'op' if $(a \text{ op } b) \in S \forall a, b \in S$ i.e. op is closure operation.

Ex $\rightarrow$ ① $S = \{1, -1\}$

op $\rightarrow *$

$(S, *) = \{(1 * -1) = -1 \in S\}$

or $(1 * 1)$ or $(-1 * 1) \in S$

So, $(S, *)$ is algebraic structure.

② $S = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

op $\rightarrow \cup$

$(S, \cup)$ is also a algebraic structure.

You can perform union in between any two elements of S and you will get the o/p lies within S.

③ $A = \{1, 2, 3\}$

$R \rightarrow$ Reflexive relation

$(R, \cup)$ and $(R, \cap)$ are algebraic structures

④ $(R, +)$ is closed under addition.

So, set can be finite or infinite

$\rightarrow$ ~~Algebraic structure also called as abstract structure~~

⑤ $S = \{1, 2, 3\}$

then, $(S, +)$ is not an algebraic structure.

Semi-Group : An algebraic structure $(S, *)$ is called a semi group if $(a * b) * c = a * (b * c) \forall a, b, c \in S$ if $'*'$ is associative on S.

Ex - ① $(N, +) \rightarrow a + (b + c) = (a + b) + c$
so, it is a semi group.

② $(N, *) \rightarrow a * (b * c) = (a * b) * c$
So, it is a semi group.

③ $(Z, -) \rightarrow$ it is only algebraic str. but not semi group.

④ $(Q^*, +) \rightarrow$ it is not algebraic str.

$Q \rightarrow$ set of rational no.

$Q^* \rightarrow Q - \{0\}$

Ex $\rightarrow (2 + (-2)) = 0$
but Q^* doesn't include 0.

⑤ $(Q^*, *) \rightarrow$ it is also an semi-group.

⑥ $(P(A), \cup), (P(A), \cap)$ both are semi-group.

Monoid ! A semigroup $(S, *)$ is called a monoid if there exist an element $e \in S$ such that $(a * e) = (e * a) = a \forall a \in S$.

$e \rightarrow$ identity element of S w.r.to operation $(*)$.

① $(N, +) : a * e = a$ [so, $e = 1$]
 $\rightarrow$ it is monoid for $e = 1$ bcoz $e \in N$.

② $(N, +) : a + e = a$ [so, $e = 0$]
but $0 \notin N$. So it is not a monoid.

③ $(Z, +) : a + e = a$ [so, $e = 0$]
 $\rightarrow$ it is monoid for $e = 0$.

Before finding 'e' verify that it should be semi-group.

④ $(P(A), \cup) : a \cup e = a$
 so $\boxed{e = \phi, a}$ but we choose
 $e = \phi$. so it is monoid.

Groups: A monoid $(S, *)$ with
 identity element 'e' is called a
 group if to each element $a \in S$,
 there exist an element $b \in S$, such
 that $(a * b) = (b * a) = e$.
 then 'b' is inverse of an element
 'a', denoted by a^{-1} .

Note →

Algebraic str. → Closure property
 ↓
 Semi-group → Associativity
 ↓
 Monoid → Identity element
 ↓
 Group → Inverse element.

Ex-① $(\mathbb{Z}, +) : \text{here } e = 0$

$$a + b = b + a = e$$

$$a + b = b + a = 0$$

$$\Rightarrow \underline{b = -a} \in \mathbb{Z}. \forall a \in \mathbb{Z}$$

so it is a group.

② $(\mathbb{Q}, *) : e = 1$

$$a * b = b * a = 1$$

$$\boxed{b = 1/a} \text{ but } \mathbb{Q} \text{ includes}$$

0 and $\frac{1}{0} = \infty$ doesn't belong to

$\mathbb{Q}$. so it is not a group.

③ $(\mathbb{Q}^*, *) \rightarrow$ it is a group

$$\text{with } \boxed{b = 1/a \quad \forall a \in \mathbb{Q}}$$

④ $(P(A), \cup) : \text{Not a group}$

$$a \cup b = b \cup a = \phi$$

we can't find any 'b' for this
 condition.

Abelian Group

→ In a group $(G, *)$, the
 following properties must
 hold good.

① Identity element and inverse
 element of 'G' is unique

② Inverse of 'e' is 'e' itself.

③ Cancellation law can be
 applied.

$$(a * b) = (a * c) \mid (a * c) = (b * c) \\ \Rightarrow \underline{b = c} \mid \Rightarrow a = b$$

$$\textcircled{4} (a * b)^{-1} = b^{-1} * a^{-1} \quad \forall a, b \in G$$

A group $(G, *)$ is abelian

if $(a * b) = (b * a) \quad \forall a, b \in G$.

It is also called commutative
group.

$$\underline{\text{Ex} \rightarrow (\mathbb{Z}, +), (\mathbb{R}^*, *),}$$

$$(\mathbb{M}, *)$$

$\mathbb{M} \rightarrow$ Set of Non-singular matrices
 of $(n \times n)$ size

Algebraic structure and also

$$(AB)C = A(BC) \text{ and also}$$

$$e = I \text{ (Identity matrix)}$$

It also have inverse $b \in \mathbb{R}$ or

non-singular so it is a group.

but $AB \neq BA$. so it is not
 abelian group.

① In a group $(G, *)$ with identity 'e', if $a * a = a$ then $a = e$.

$$\Rightarrow a * a = a$$

$$\Rightarrow a * a = a * e$$

$$\Rightarrow [e = a] \rightarrow \text{True}$$

② In a group $(G, *)$ if $x^{-1} = x \forall x \in G$ then G is abelian group.

$$(a * b)^{-1} = b^{-1} * a^{-1}$$

$$(a * b) = b * a \rightarrow \text{True}$$

$\rightarrow$ So it is abelian group

③ In group $(G, *)$ if

$$(a * b)^2 = a^2 * b^2 \forall a, b \in G \text{ then } G \text{ is abelian group.}$$

$$\Rightarrow (a * b)^2 = a^2 * b^2$$

$$(a * b) * (a * b) = a * a * b * b$$

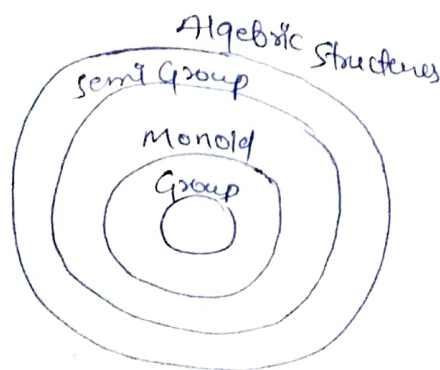
$$a * (b * a) * b = a * (a * b) * b$$

$$\Rightarrow a * b = b * a$$

So it is abelian.

$\rightarrow$ 'e' is not multiplication, it can be any operation.

Ex-4



Q Let $A = \{x \mid 0 < x \leq 1 \text{ and } x \in \mathbb{R}\}$ then 'A' w.r.to multiplication is -

- a) a semigroup but not monoid
- b) monoid but not a group
- c) a group
- d) Not a semi-group.

$$x \in (0, 1] = A$$

$A \rightarrow$ algebraic str. and semi group.

$e = 1$ here so monoid.

$b = 1/a \notin A$ so, inverse existence is not possible.

So, it is not a group.

b) monoid but not a group

Q Let A is set of all integers and a binary operation $+$ is defined by $(a + b) = \min(a, b)$, then $(A, +)$ is -

① Algebraic structure ✓

② Semi group: It also follows associativity so, semigroup

③ Monoid: $a + e = a$

We can't find a finite e , so that it works as identity. (e can be ∞)

$\rightarrow (A, +)$ is a semi group only.

Q Let S be the string set of all bits accepts the null strings ϵ .

'+' denotes string concatenation. $(S, +)$ is.

$$S = \{\epsilon, 0, 1, 00, 11, 01, 001, \dots\}$$

Set is algebraic structure for $(S, +)$

$$(a + b) + c = a + (b + c)$$

$$abc = abc$$

- So it is semi group
- Identity element $e = \epsilon$ here. So its monoid.
- $a + e = \epsilon$, $\exists$ is never possible. So, $(S, +)$ is monoid only, not a group.

Not → for identity element

$$\begin{array}{l} a * e = a \\ \textcircled{2} \text{ for inverse element } a * a^{-1} = e \end{array} \quad \left| \begin{array}{l} * \rightarrow \exists + \\ \text{can be any operation} \end{array} \right.$$

Finite Group → Group with finite no. of elements.

OCG → Order of finite group means no. of elements present in group.

Ex → $(\{0\}, +) \rightarrow \exists$ is AS, semi-group, $e = 0$ (so its monoid).
 $\Rightarrow a^{-1} = 0$ so its group.

→ $\exists$ is only finite group over real no's containing one element over '+' operator.

$(\{1\}, *) \rightarrow \exists$ is also a finite group.

Only a finite group over '+' for real no's with one element

$$(\{1, -1\}, *) \quad \begin{array}{c|cc} & 1 & -1 \\ \hline 1 & 1 & -1 \\ -1 & -1 & 1 \end{array}$$

All element in composition table lies in set, so it is algebraic structure

* → $\exists$ is associative

$e = 1$ (Identity element)

→ $\exists$ also have inverse as for 1 inverse is 1 and for -1 inverse is -1. So $\exists$ is a finite group.

$$(\{1, \omega, \omega^2\}, *) : \omega^3 = 1$$

	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

So, it is alg. structure.

* is associative, so it is semi-group.

$e = 1$ here as identity element
 Inverse of 1 → 1,
 $\omega \rightarrow \omega^2, \omega^2 \rightarrow \omega$. So for all element inverse exist. So its a group.

→ Above is cube root of unity.

$$(\{0, 1, 2, \dots, (m-1)\}, \oplus_m)$$

$\oplus_m \rightarrow$ Addition modulo m.
 $\exists$ means first sum and the find % m.

$$\text{If } (a+b) < m \text{ then o/p} = (a+b) \\ (a+b) > m \text{ then o/p} = (a+b) \% m$$

$$1 \oplus_4 2 = 3 \% 4 = 3$$

$$3 \oplus_4 4 = 7 \% 4 = 3$$

$\exists$ modulo 'm' o/p range is always $[0 \rightarrow (m-1)]$

→ Above set is a group bcoz it has both identity and inverse bcoz range is upto (m-1). Hence it

if we also include 'm' then we will find 'no' inverse for 'm'. So it will not be a group with $S = \{0, 1, \dots, m\}$.

→ for multiplication modulo m , '1' is identity element generally.

→ for $(\oplus)_m$, '0' is an identity element generally.

Order → It defines an element of group (finite group).

Let $(G, *)$ be a group, and $a \in G$, then order of element 'a' is the smallest +ve integer n such that a^n is identity element.

Ex → $(\{1, -1\}, *)$

Here, $e = 1$ (identity element)

$$\text{So, } (1)^n = e \Rightarrow (1)^n = 1$$

So, $n=1$ [Smallest]

$$(-1)^n = e = 1$$

$$\Rightarrow n=2 \text{ [Smallest]}$$

$$\underline{O(-1) = 2}, \quad \underline{O(1) = 1}.$$

Order of Identity element is always 1. $O(1) = 1$.

$$(\{1, \omega, \omega^2\}, +)$$

Here $e = 1$

$$\text{So, } O(1) = 1 \quad | \quad O(\omega^2) = 3$$

$$O(\omega) = 3$$

$$(\omega^3) = 1 = \omega^6$$

$O(\text{Group}) = \text{no. of elements present in group.}$

Order of element always divides order of a group

$$(\{0, 1, 2\}, \oplus_3) \quad O(G) = 3$$

$$I = 0$$

$$0 \oplus 0 = 0 \quad O(0) = 1$$

$$1 \oplus_3 1 \oplus_3 1 = 3 \% 3 = 0$$

$$2 \oplus_3 2 \oplus_3 2 = 6 \% 3 = 0$$

$$\underline{O(1) = O(2) = 3.}$$

→ Order of element a and a^{-1} is going to be same

$O(\omega) = O(\omega^2) = 3$ bcoz both are inverse of each other

Subgroups

Let $(G, *)$ be a group. A subset H of 'G' is called a subgroup of 'G' if $(H, *)$ is a group.

① → A group is subgroup of itself

② → Group with set containing only identity elements are also called subgroup.

→ In above ② statement subgroups defined are called as trivial subgroups.

→ All subgroup other than this called proper subgroup.

Ex → $G = (\{1, -1, i, -i\}, *)$
 then, $H = (\{1, i\}, *)$ is
 a proper subgroup.
 $(\{1\}, *) \rightarrow$ Trivial Subgroup
 and G

Theorems ($+$ → Any operation)

① Let H be a non-empty subset
 of a group $(G, +)$. H is a
 subgroup of ' G ' iff $a+b \in H$
 $\forall a, b \in H$.

→ It means if you choose any pair
 of (a, b) from H , it should satisfy
 the property $a+b \in H$.

② Let H be a non-empty finite
 subset of a group $(G, +)$. H
 is a subgroup of ' G ' iff
 $(a+b) \in H \forall a, b \in H$

③ Lagrange's Theorem

If H is a subgroup of finite
 group $(G, +)$, then $O(H)$ is the
 divisor of $O(G)$.

The converse of Algo. need not be
 true.

$O(G) = m$ $O(H) = n$ then m/n
 should exist

→ Using this we can directly
 discard some sets.

Ex Say $O(G) = 10$, $O(H) = 3$
 then H will not be subgroup
 but it can be subset

→ But if divides then only
 we check for H is subgroup
 or not.

→ It is a kind of (-ve) test.

Ex → $G = (\{0, 1, 2, 3, 4, 5\}, (+)_6)$

Which of the following is
 subgroup of G .

① $H_1 = \{1, 3\}$

$$O(G) = 6 \quad O(H_1) = 2$$

$6/2 = 3$. So H_1 can be group

Now apply theorem ②

	1	3
1	2	4
3		

doesn't belong
 to H_1

So, it doesn't follow
 closure property hence it
 is not a subgroup.

② $H_2 = \{1, 5\}$

$$O(G)/O(H) = 3$$

	1	5
1	2	0
5		

2 is not present
 in H_2

③ $H_3 = \{0, 3\}$

$$O(G)/O(H) = 3$$

	0	3
0	0	3
3	3	0

So all the
 elements are
 present

So it is a subgroup

④ $H_4 = \{0, 2, 4\}$

$$O(G)/O(H_4) = 2$$

	0	2	4
0	0	2	4
2	2	4	0
4	4	0	2

→ It is
 a
 sub-
 group

(c) $H_5 = \langle 0, 2, 3, 5 \rangle$

$|O(H)|/|O(H_5)| = 3/2 = 1.5$

So, it is not a subgroup.

→ Let $(G, *)$ be a group of order P where P is a prime no., then the no. of proper subgroups of G is 0.

$d(G) = P$. (Given)

$O(H)$ can be $\{1, P\}$ b/c prime no. is only divisible by 1 and itself.

→ $O(H) = 1$ and P both denotes trivial subgroup.

So, no. of proper subgroup is 0.

→ $(S_8, \oplus_8) = (\{0, 1, 2, 3, \dots, 7\}, \oplus_8)$

Which of the following are not true

① The union of two subgroups of ' G ' is also a subgroup of ' G '. → false

Ex → $(S_8, \otimes_8)$
 $= (\{1, 3, 5, 7\}, \otimes_8)$

So, $H_1 = (\{1, 3\}, \otimes_8)$ $H_2 = (\{1, 5\}, \otimes_8)$

1	3
1	3
3	1

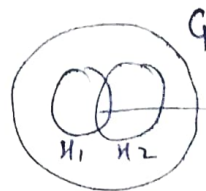
Both are subgroups

$H_1 \cup H_2 = \{1, 3, 5\}$

$H_4 \quad 3 * 5 = 15 \% 8 = 7$

but 7 doesn't $\in H_1 \cup H_2$
 so it is false statement

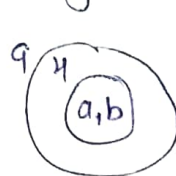
② The intersection of two subgroups of ' G ' is always a subgroup of G . → True



all elements in this area should satisfy closure property

③ The union of two subgroups H_1 and H_2 is also a subgroup → false

④ Every subgroup of an abelian group is also an abelian group. → True



$a * b = b * a \quad \forall a, b \in H$

Cyclic Groups : A group

$(G, *)$ is called as a cyclic group if there exists an element $a \in G$ such that element in ' G ' can be written as a^n for some integer ' n '. Then ' a ' is called generator or generating element

Ex → $G = (\{1, -1\}, +)$

$(-1)^2 = 1 \quad (-1)' = -1$

$G = (\{1, \omega, \omega^2\}, +)$

$(\omega)' = \omega, (\omega)^2 = \omega^2$

$(\omega)^3 = 1$

ω^2, ω → generating elements

find out all the generator of Group.

Ex $\rightarrow G = (\mathbb{Z}_{1,2,3,4,5,6}, \oplus_7)$

$\rightarrow$ No. of elements that can be works as generator is 'n'. To compute n find $\phi(6) = S_6 = \{1, 5\}$

$S_6 \rightarrow$ Elements less than 6 and relatively prime.

So, $n=2$ here (if it is cyclic)

Now find the generating element

On doing some calculation we will get that 3 is generator. So, it is cyclic, hence we can say that,

$$\begin{array}{ccc} 3^1 \% 7 & \text{and} & 3^5 \% 7 \\ \downarrow & & \downarrow \\ 3 & & 5 \end{array}$$

So, 3, 5 are two generators.

Theorems

Note:-

Two numbers are relatively prime if they nothing in common factor other than 1.

In $\mathbb{Z}_{1,2,3,4,5,6}$

factor of	factors	Relatively prime to
1	1	
2	1, 2	$6 = \{1, 5\}$
3	1, 3	bcuz 2 and 6 have 2 as common and 3 and 6 have 3 as common
4	1, 2, 4	
5	1, 5	
6	1, 2, 3, 6	

Similarly relatively prime to 5 = $\{1, 2, 3, 4\}$

bcuz, 1, 2, 3, 4 have no common element other than 1.

Theorem 1:

Let $(G, +)$ be a cyclic group of order 'n' with generator 'a', then

(i) The number of generators is $\phi(n)$

$\phi(n) \rightarrow$ Euler's function of 'n'

(ii) a^m is also a generator if $\text{gcd}(m, n) = 1$

$\phi(3) \rightarrow$ no. of integers less than 3 which are relatively prime to 3

$$\phi(3) = \{1, 2\} = 2$$

$$\phi(7) = \{1, 2, 3, 4, 5, 6\} = 6$$

But it is valid for cyclic group only

Ex $\rightarrow$ Let $(G, +)$ be a cyclic group of order 8 with generator 'a'.

(1) Number of generators in G is $\phi(8)$.

$$\phi(8) = S_8 = \{1, 3, 5, 7\} = 4$$

a^1, a^3, a^5, a^7 are generating

(2) which of the following is not a generator

- ① a^2 ② a^3 ③ a^5 ④ a^7

① $n = pq$ then
 $\phi(n) = \phi(p) \cdot \phi(q)$
 $\forall p, q$ distinct values and prime

② $\phi(p) = p-1 \quad \forall p \in \text{Prime nos}$

③ Ex: $\phi(77) = \phi(7) \cdot \phi(11)$
 $= 6 \cdot 10$
 $= \underline{60}$

$\phi(35) = \phi(7) \cdot \phi(5)$
 $= 6 \cdot 4$
 $= \underline{24}$

Exception:

① for $\phi(25) = \phi(5^2)$
 $= 5^2 - 5^1 = 20$

$\phi(p^n) = p^n - p^{n-1}$
 $\forall p \in \text{Prime numbers}$

→ This formula is only applicable to repetitions.

② $\phi(84) = \phi(2^2 \times 3 \times 7)$
 $= \phi(2^2) \phi(3) \phi(7)$
 $= (4-2) \times \downarrow \times \downarrow$
 $= 2 \times 2 \times 6$
 $= \underline{24}$

→ If $(G, *)$ is a cyclic group with generator 'a' then

① a^{-1} is also a generator

② The order of generator $= \phi(G)$

Ex → $(\{0, 1, 2, 3\}, \oplus_4)$

$1^1 = 1 \quad 1^2 = 2 \quad 1^3 = 3$

$1^4 = 0 \rightarrow \text{order}$

$3^1 = 3 \quad 3^2 = 2 \quad 3^3 = 1$

$3^4 = 0 \rightarrow \text{order}$

for $e=0$ (1, 3) are inverse of each other

So, property 1 follows.

$\phi(2) = \phi(3) = \phi(4) = 4$. So it also satisfy point ② but only for cyclic groups.

Properties of Cyclic Group

① Every Cyclic group is abelian group.

② Every group of prime order is cyclic and so every group of prime order is abelian group.

③ Every subgroup of a cyclic group is also cyclic, but the generator of the subgroup need not be same as that of the cyclic group.

Ex: $G = \{1, -1, i, -i\}$, $H = \{1, -1\}$
 So, H is a subgroup of G .

Generators of $G = i, -i$, generator of $H = -1$.

④ Let $(G, *)$ be a group of even order, then there exist ^{at least} one element $a \in G$ ($a \neq e$) such that $a^{-1} = a$

→ for first statement

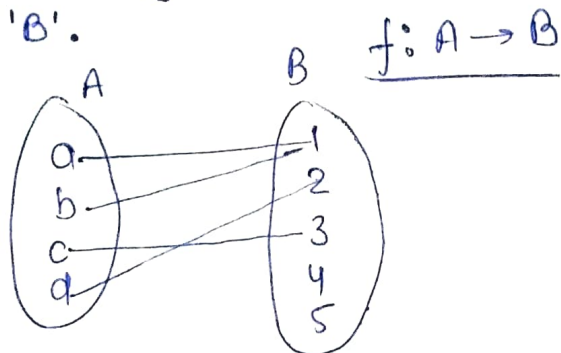
$a * b = g^n * g^m = g^{n+m} = g^m * g^n$
 $= b * a$

So every cyclic group is abelian.

$g \rightarrow$ any generator and in cyclic group we can write elements in power of generators.

functions

A relation 'f' from a set 'A' to a set 'B' is called a function if to each element $a \in A$, we can assign an unique element of 'B'.



$$R = \{(a, 1), (b, 1), (c, 3), (d, 2)\}$$

$$\text{Domain} = \{a, b, c, d\} = \underline{A}$$

$$\text{Co-domain} = \{1, 2, 3, 4, 5\} = \underline{B}$$

$$\text{Range} = \{1, 2, 3\}$$

→ 'Each element' in A must be related to B and each element in A should get only one image in B.

In $(a, 1) \rightarrow 1$ is image of a and 'a' is pre-image of 1.

→ In relation, 'B' can have more than one pre-image but 'A' can't have more than one image.

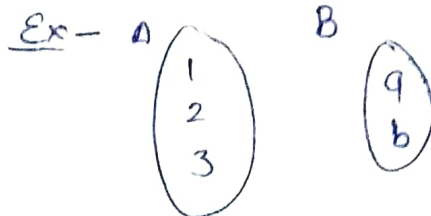
→ Range $\subseteq$ Co-Domain

Count the no's of function

$$f: A \rightarrow B$$

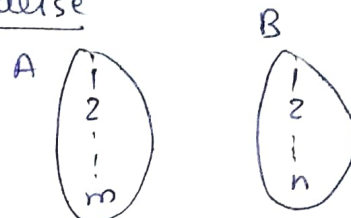
$$|A| = m$$

$$|B| = n$$



Each element in A have 2 choices to choose in B.
= No. of fn possible = $2 \times 2 \times 2 = 8$.

Generalise



Each element have n choices.

$$\text{No. of function} = \boxed{n^m} = |B|^{|A|}$$

→ Every relation is not a function.

No. of Relations possible over A and B which are not functions = $N(R)$

$$= \text{Total no. of relation} - \text{Total no. of functions.}$$

$$\text{Total no. of relation } N = 2^{mn}$$

$$\text{So, } N(R) = 2^{mn} - n^m$$

$$\text{for } f: A \rightarrow A \quad |A| = n$$

then,

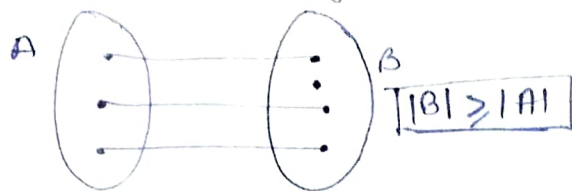
$$\text{No. of function} = n^n$$

$$\text{No. of relations} = 2^{n^2}$$

$$N(R) = 2^{n^2} - n^n$$

One-one function

→ Also called Injection.



Each element in A have distinct image in B.

→ If there are exactly one-one functions possible from $A \rightarrow B$ then which of following is not true

a) $|A| = 5$ & $|B| = 5 \rightarrow 5P_5$

b) $|A| = 4$ & $|B| = 5 \rightarrow 5P_4$

c) $|A| = 3$ & $|B| = 6 \rightarrow 6P_3$

d) $|A| = 5$ & $|B| = 4 \rightarrow \underline{X}$

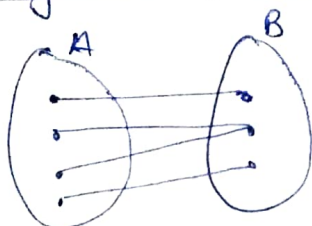
No. of one-one fn = $m \cdot (m-1) \cdot (m-2) \dots (m-n+1)$
 $= \boxed{mP_n}$

$|A| = n$, $|B| = m$

Onto functions (Surjective)

A function $f: A \rightarrow B$ is said to be onto if each element of B is mapped to atleast one element in A.

Range = Co-Domain



$|B| \leq |A|$ → Necessary Condition, not sufficient condition.

→ function that shows both the properties injective and surjective called bijective.

Bijective → one-one & onto

→ If one-one function have $|A| = |B|$ then it is onto.

→ If $|A| = |B| = n$ then no. of onto functions = $\underline{n!}$ and no. of one-one function = $\underline{n!}$

→ If $|A| = m$ and $|B| = n$, ($m > n$) then the no. of onto functions possible from $A \rightarrow B$ is,

$$n^m - nC_1(n-1)^m + nC_2(n-2)^m - \dots + (-1)^{n-1} nC_{n-1} (1)^m$$

Ex: $|A| = 6$ $|B| = 3$, no. of onto functions from $A \rightarrow B$.

$m = 6$, $n = 3$

$$3^6 - 3C_1(2)^6 + 3C_2(1)^6 - 3C_3(0)^6 = \underline{540}$$

Ex → In how many ways we can assign 5 employees to 4 project so that every employee is assigned to only one project and every project is assigned to atleast one employee.

$m = 5$ $n = 4$

e_1
 e_2
 e_3
 e_4
 e_5

p_1
 p_2
 p_3
 p_4

required ways =

$$4^5 - 4C_1 3^5 + 4C_2 2^5 - 4C_3 1^5 + 0 = \underline{240}$$